JAYPEE UNIVERSITY
ANOOPSHAHR
UTTAR PRADESH

BRAINYCAMPUS
SMART ACADEMIC MANAGEMENT SYSTEM

PROJECT REPORT

Submitted in partial fulfillment of the requirement for the award of degree of

MASTER OF COMPUTER APPLICATIONS (MCA)

Submitted By

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Under The Guidance Of

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Academic Session: 2025–2026

CERTIFICATE

This is to certify that the project entitled "Brainy Campus – Smart Academic Management System" submitted by Srishti (Roll No. 8825501002) and Kajal Kumari (Roll No. 8825501013) in partial fulfillment of the requirements for the award of the degree of Master of Computer Applications (MCA) has been completed successfully under my guidance.

Guide Signature	HOD Signature
Mr. Rahul Gupta	Head of Department

Date:
Place:

DECLARATION

We hereby declare that the project report entitled “Brainy Campus – Smart Academic Management System” submitted by us is an original work carried out under the guidance of Mr. Rahul Gupta.

This project has not been submitted previously for any degree or diploma.

Signature
Srishti
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ACKNOWLEDGEMENT

We express our sincere gratitude to our project guide Mr. Rahul Gupta for his valuable guidance and support throughout the development of this project.

We also thank the Department of MCA, JP University Anoopshahr for providing us the opportunity and resources required for completing this project successfully.

We are thankful to our friends and family members for their encouragement and support.

ABSTRACT

Brainy Campus is a Smart Academic Management System developed using Django framework with SQLite database. The system provides a centralized platform for students to manage attendance, assignments, notes, timetable, reminders, and academic activities efficiently.

The project helps improve productivity and reduces manual academic management problems through a user-friendly dashboard. The frontend of the system is developed using HTML, CSS, and JavaScript while backend processing is handled through Django.

The system improves accessibility, organization, and academic tracking for students and teachers.

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Preliminary Pages

CHAPTER 1 – INTRODUCTION

BrainyCampus – Smart Academic Management System is a web-based application developed to simplify and digitize academic management for students and educational institutions. In traditional academic systems, students often face difficulties in managing attendance, assignments, notes, timetables, and exam schedules due to scattered and manual processes. This project aims to overcome these challenges by providing a centralized digital platform where all academic activities can be managed efficiently in one place.

The BrainyCampus system is designed using the Django framework with Python as the backend language and HTML, CSS, and JavaScript for the frontend interface. SQLite is used as the database for storing and managing academic data securely. The system follows a structured and modular approach, making it easy to use, maintain, and upgrade.

This application provides role-based access for students and administrators, ensuring secure login and proper data management. Each user can access personalized dashboards where they can manage attendance records, track assignments, store notes, plan exams, view timetables, and stay updated with academic events. The interface is designed to be simple, responsive, and user-friendly so that users can easily navigate through different modules without confusion.

The main aim of this project is to improve productivity, reduce manual workload, and enhance academic organization. By integrating all essential academic tools into a single platform, BrainyCampus helps students manage their studies more effectively and allows institutions to maintain better academic records digitally.

In conclusion, BrainyCampus acts as a smart solution for modern academic management by combining automation, accessibility, and efficiency into one system, making academic processes more structured and reliable.

CHAPTER 2 – PROBLEM STATEMENT

In traditional educational systems, academic management is mostly handled using manual methods or multiple disconnected tools. Students and teachers often rely on paper-based records, spreadsheets, and separate applications for managing attendance, assignments, notes, timetables, and examination schedules. This fragmented approach creates several difficulties in maintaining proper academic organization and efficiency.

One of the major problems is the lack of a centralized platform for academic activities. Students are required to check different sources for assignments, notices, and study materials, which leads to confusion and mismanagement of important academic tasks. Similarly, teachers face challenges in maintaining accurate attendance records and tracking student performance manually, which is time-consuming and prone to human errors.

Another significant issue is the difficulty in tracking deadlines and academic schedules. Students often miss assignment submissions and exam preparations due to poor organization and absence of automated reminders. There is also no unified system that provides real-time updates regarding academic progress, attendance percentage, or upcoming events.

Furthermore, traditional systems do not provide secure and structured data management. Important academic information may get lost, mismanaged, or become difficult to retrieve when needed. The lack of integration between different academic functions also reduces productivity and increases administrative workload.

Therefore, there is a strong need for a centralized, automated, and user-friendly academic management system that can integrate all academic activities into a single platform. The BrainyCampus – Smart Academic Management System is designed to address these issues by providing a digital solution that improves organization, efficiency, and accessibility for both students and faculty members.

CHAPTER 3 – OBJECTIVES OF THE PROJECT

The main objective of the **BrainyCampus – Smart Academic Management System** is to design and develop a centralized digital platform that simplifies and improves the management of academic activities for students and educational institutions. The system aims to replace traditional manual processes with an efficient, organized, and user-friendly online solution.

One of the primary objectives of this project is to provide a single platform where all academic activities such as attendance management, assignment tracking, timetable scheduling, notes management, exam planning, and event updates can be accessed easily. This helps in reducing dependency on multiple tools and ensures better organization of academic data.

Another important objective is to improve productivity and time management for students. By providing structured modules and timely information, the system helps students plan their studies effectively, track deadlines, and monitor their academic progress without confusion.

The system also aims to reduce manual workload for teachers and administrative staff. Automated features such as attendance calculation, assignment tracking, and data storage minimize human errors and save valuable time that can be utilized for teaching and academic improvement.

Security and data management is another key objective of the project. The system ensures that all user data is stored securely in the database with proper authentication mechanisms. Only authorized users such as students and faculty members can access their respective data, ensuring privacy and data protection.

Additionally, the project aims to provide a simple and user-friendly interface so that users with minimal technical knowledge can easily navigate and use the system without difficulty. The design focuses on clarity, responsiveness, and ease of access.

In conclusion, the BrainyCampus system is developed with the objective of creating a smart, centralized, and efficient academic management solution that enhances the overall learning experience and improves academic organization in educational institutions.

CHAPTER 4 – EXISTING SYSTEM

In the existing academic management system, most educational institutions rely on traditional and semi-digital methods for handling academic activities. These methods include paper-based records, spreadsheets, and the use of multiple independent applications for different tasks such as attendance, assignments, notes, and timetable management.

In the traditional system, attendance is recorded manually by faculty members in registers or basic spreadsheets. This process is time-consuming and increases the chances of human errors, miscalculations, and data inconsistency. Similarly, assignment distribution and submission are managed through classroom announcements, printed notices, or messaging applications, which often leads to missed deadlines and poor tracking.

Students also face difficulties in organizing their academic activities because there is no centralized platform that integrates all essential functions. Important information such as exam schedules, study materials, and event updates are scattered across different sources, making it difficult for students to manage their academic responsibilities efficiently.

Moreover, manual systems lack real-time updates and automated reminders. This results in poor communication between students and faculty members and reduces overall academic productivity. Data retrieval is also difficult in case of physical records, and there is always a risk of data loss or damage.

Therefore, the existing system is inefficient, time-consuming, and lacks proper integration of academic modules. These limitations highlight the need for a modern, centralized, and automated solution like the BrainyCampus system.

COMPARISON TABLE

Feature	Existing System	BrainyCampus System
Data Storage	Manual / Paper-based	Digital Database (SQLite)
Attendance Tracking	Registers / Excel	Automated System
Assignment Management	Notices / WhatsApp	Centralized Platform
Accessibility	Limited	Anytime & Anywhere
Data Security	Low	High with Authentication
Efficiency	Low	High
Error Rate	High	Minimal

CHAPTER 5 – PROPOSED SYSTEM

The proposed system, **BrainyCampus – Smart Academic Management System**, is a modern and centralized web-based solution designed to overcome the limitations of the existing manual academic management methods. It provides an integrated platform where all academic activities are managed digitally in a structured and efficient manner.

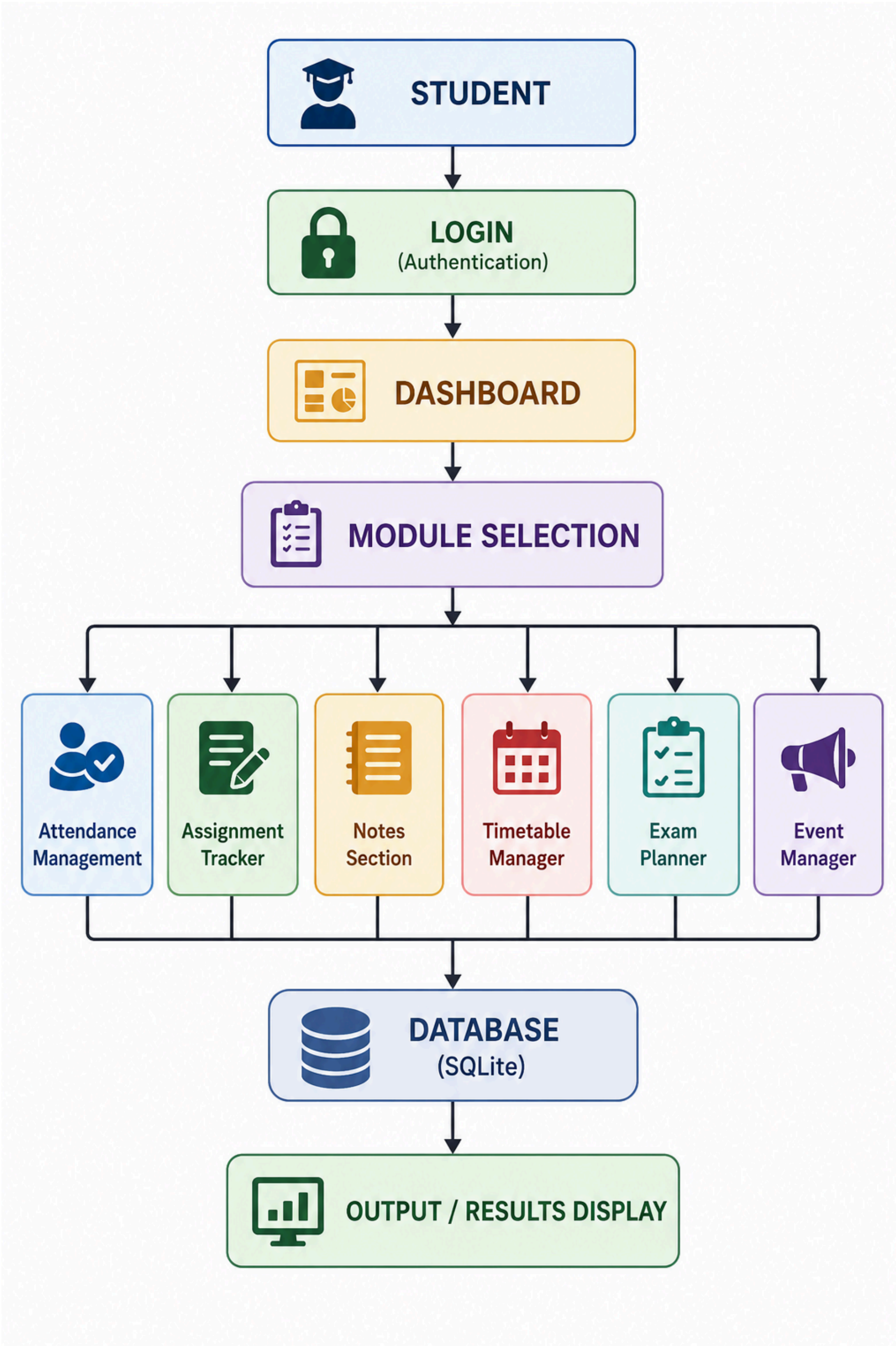
The system is developed to provide a user-friendly interface that allows students and faculty members to access academic information easily. It includes essential modules such as attendance management, assignment tracking, notes management, timetable scheduling, exam planning, and event management. All these modules are integrated into a single dashboard to ensure smooth academic operations.

The proposed system uses Django as the backend framework, SQLite as the database, and HTML, CSS, and JavaScript for the frontend design. It also includes secure authentication to ensure that only authorized users can access the system. This improves data security and privacy.

One of the key features of the proposed system is automation. Tasks such as attendance calculation, assignment tracking, and reminder notifications are handled automatically, ide a smart dreducing manual effort and improving accuracy. The system also provides better communication between students and faculty by centralizing all academic data.

Overall, the proposed system aims to improve efficiency, reduce paperwork, enhance productivity, and provigital solution for academic management.

❖ BASIC SYSTEM FLOW DIAGRAM :



CHAPTER 6 – TECHNOLOGY STACK

The Brainy Campus – Smart Academic Management System is developed using a modern and efficient technology stack. These technologies ensure smooth performance, scalability, and an interactive user experience for students and faculty members.

✔ Frontend Technologies

The frontend is responsible for the user interface and design of the system.

- **HTML (HyperText Markup Language)**
  Used to create the structure of web pages such as login, dashboard, and module pages.
- **CSS (Cascading Style Sheets)**
  Used to design and style the application, making it visually appealing and user-friendly.

✔ Backend Technology

Django (Python Framework)

Django is used as the backend framework of the system. It handles:

- User authentication
- Database operations
- Business logic
- URL routing and views

It follows the MVT (Model–View–Template) architecture, which makes development structured and maintainable.

- **Python**
  Python is the core programming language used to implement backend logic and integrate all system modules.

✔ Database

- **SQLite3**
  SQLite is used as the database system for storing all academic data such as:
  - Student information
  - Attendance records
  - Assignments

- Notes
- Timetable data
- Notebook data
- Events Manager

It is lightweight, fast, and suitable for academic-level projects.

CHAPTER 7 – SYSTEM ARCHITECTURE

✔ System Architecture Explanation

The Brainy Campus – Smart Academic Management System follows a **three-tier architecture** that separates the system into three main layers: Frontend, Backend, and Database. This structure improves maintainability, scalability, and performance of the application.

🔹 1. Frontend Layer

The frontend layer is the user interface of the system. It is developed using **HTML, CSS, and JavaScript**.
This layer is responsible for:

- Displaying web pages to the user
- Collecting user input
- Providing a simple and interactive interface

Users interact with the system through the frontend (login page, dashboard, modules, etc.).

🔹 2. Backend Layer (Django)

The backend is developed using the **Django framework with Python**.
It acts as the core processing unit of the system.
Functions of backend:

- Handles user authentication and authorization
- Processes user requests
- Implements business logic
- Communicates between frontend and database
- Manages CRUD operations (Create, Read, Update, Delete)

🔹 3. Database Layer (SQLite)

The database layer stores all the academic data using **SQLite3**.
It stores:

- Student details
- Attendance records
- Assignments
- Notes
- Timetable data
- Events

SQLite ensures lightweight, fast, and reliable data storage.

☐ SYSTEM ARCHITECTURE DIAGRAM:

CHAPTER 8 – DATABASE DESIGN

The Brainy Campus – Smart Academic Management System uses **SQLite3** as its database. The database is designed in a structured and relational format to store all academic information securely and efficiently. Each module of the system is connected through the **User table**, which acts as the central entity.

✔ Database Overview

The database stores all academic data such as attendance records, assignments, notes, timetable entries, and events. It ensures data integrity, fast retrieval, and organized storage for all users.

Tables in the Database

🔹 1. User Table

This is the main table of the system.
Purpose:
Stores user login and profile information.
Fields:

- User ID
- Username
- Password
- Email
- Role (Student / Admin)

🔹 2. Attendance Table

Stores attendance records of students.
Fields:

- User (Foreign Key)
- Subject
- Total Classes
- Attended Classes
- Attendance Percentage

🔹 3. Assignment Table

Manages student assignments.
Fields:

- User (Foreign Key)
- Title
- Due Date
- Status (Completed / Pending)

🔹 4. Notes Table

Stores study notes created by users.
Fields:

- User (Foreign Key)
- Title
- Content
- Created Date

🔹 5. Timetable Table

Stores class schedules.
Fields:

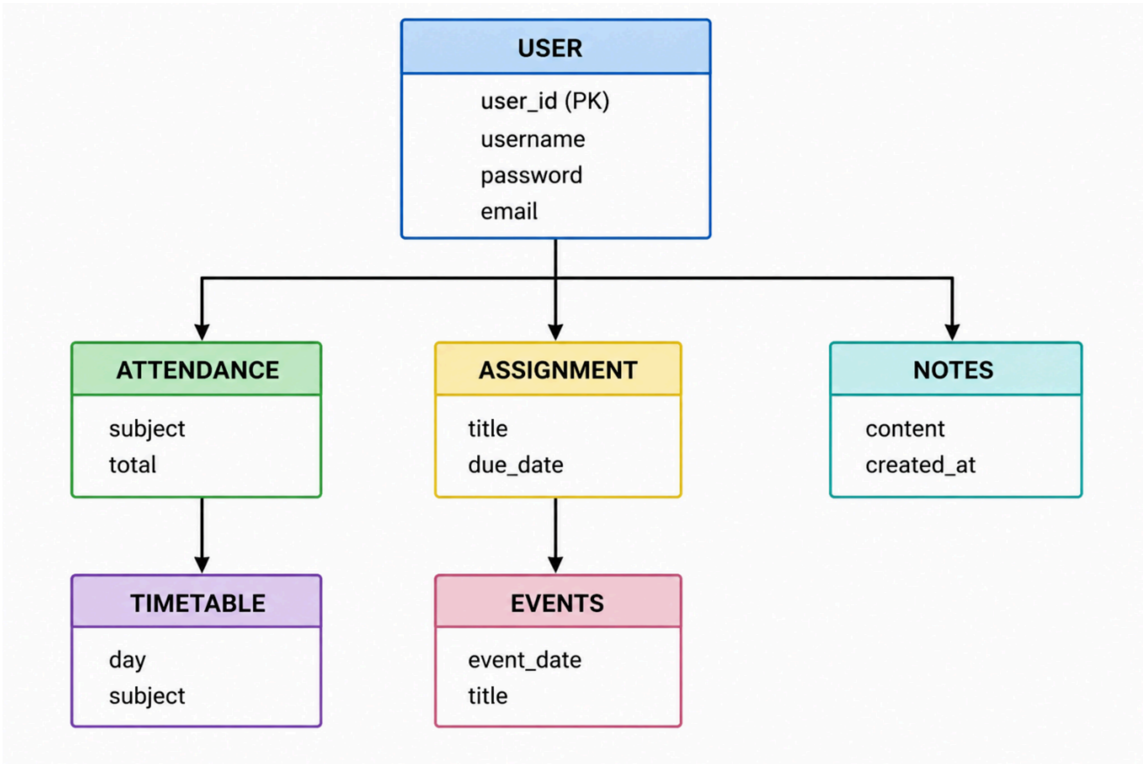
- User (Foreign Key)
- Day
- Subject
- Time

🔹 6. Events Table

Stores academic events and important dates.
Fields:

- User (Foreign Key)
- Event Title
- Event Date

☐ ER DIAGRAM :


USER
user_id (PK)
username
password
email
ATTENDANCE
subject
total
ASSIGNMENT
title
due_date
NOTES
content
created_at
TIMETABLE
day
subject
EVENTS
event_date
title

CHAPTER 9 – MODULES OF THE SYSTEM

The Brainy Campus – Smart Academic Management System is divided into multiple functional modules. Each module is designed to handle a specific academic task in a structured and efficient manner. These modules work together to provide a complete academic management solution for students and faculty members.

✔ Core Modules of the System

1. Timetable Manager

The Timetable Manager helps students organize their daily class schedule efficiently.
Features:

- Add and manage class timings
- View day-wise timetable
- Organize subjects and lectures
- Avoid scheduling conflicts

2. Assignment Tracker

This module is used to manage academic assignments.
Features:

- Track assignments and deadlines
- Mark assignment status (Completed/Pending)
- Organize assignments subject-wise
- Improve submission management

3. Attendance Module

The Attendance Module helps track student attendance records.
Features:

- Record attendance subject-wise
- Calculate attendance percentage
- Identify low attendance alerts
- Maintain attendance history

4. Notes Module

This module allows students to store and manage study notes.
Features:

- Create and save notes
- View and edit notes anytime
- Organize subject-wise notes
- Easy access from dashboard

5. Notebook Module

The Notebook module is used to store detailed study materials.
Features:

- Store long study content
- Organize academic materials
- Easy retrieval of notes
- Improve learning structure

6. Exam Planner

This module helps students plan and manage exams effectively.
Features:

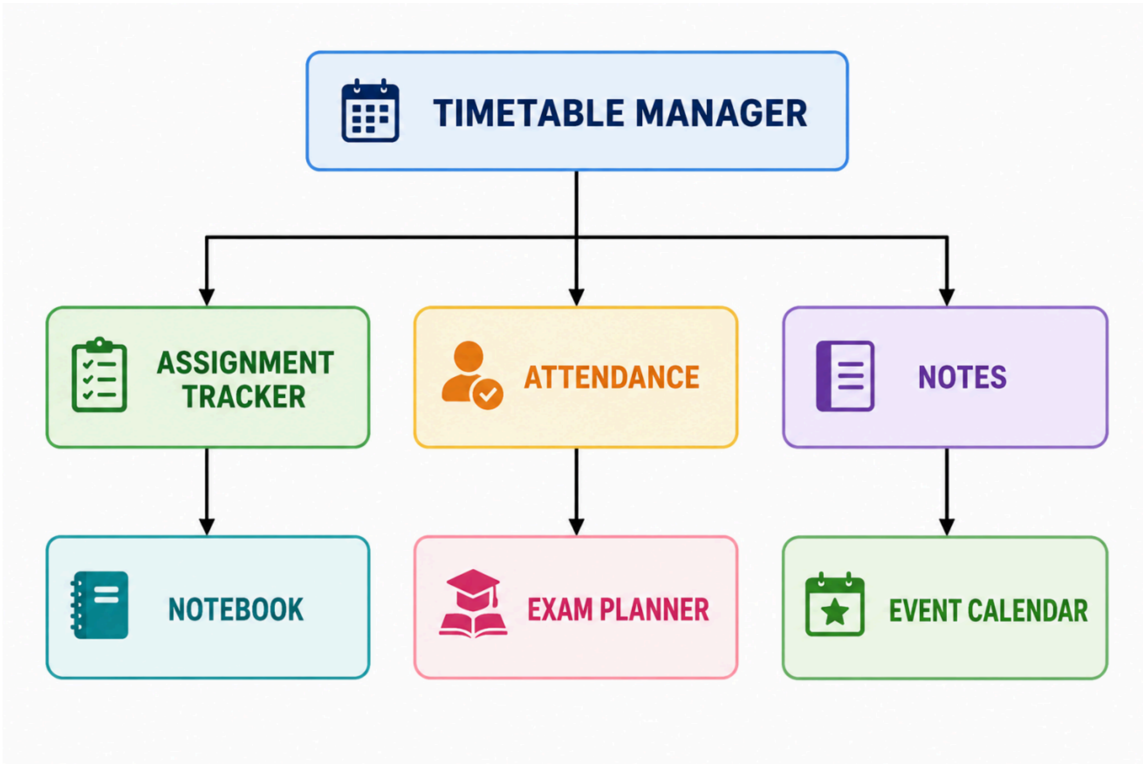
- Add exam schedules
- Track exam dates
- Plan preparation strategy
- Improve exam time management

7. Event Calendar

The Event Calendar module tracks important academic events.
Features:

- Manage academic events
- Track important dates
- Organize college activities
- Provide event reminders

☐ BLOCK DIAGRAM OF MODULES :



CHAPTER 10 – BACKEND FUNCTIONALITY

The backend of the Brainy Campus – Smart Academic Management System is developed using the **Django framework with Python**. It plays a crucial role in handling the core logic of the application, processing user requests, managing data, and ensuring secure communication between the frontend and the database.
Django follows the **MVT (Model–View–Template) architecture**, which helps in maintaining a clean separation of concerns and makes the application easy to develop, scale, and maintain.

✔ 1. User Authentication System

The system provides a secure login and logout mechanism using Django's built-in authentication system.
Features:

- User registration and login
- Password authentication
- Session management
- Role-based access (Student / Admin)

Only authenticated users are allowed to access the dashboard and modules.

✔ 2. Views and Request Handling

Django views are used to handle all HTTP requests from the user.
Functions of Views:

- Process user input from forms
- Validate data
- Return appropriate HTML pages
- Redirect users based on actions (login, logout, home page)

Example:

- Login view handles authentication
- Home view displays dashboard after login

✔ 3. Models and Database Interaction

Django models are used to define the structure of the database.
Functions:

- Create database tables automatically
- Perform CRUD operations (Create, Read, Update, Delete)

- Maintain relationships between tables using Foreign Keys

Models used in the system include:

- User
- Attendance
- Assignment
- Notes
- Timetable
- Events

✔ 4. CRUD Operations

The backend supports full CRUD functionality for all modules.

- **Create:** Add new attendance, assignments, notes, etc.
- **Read:** View stored academic data
- **Update:** Modify existing records
- **Delete:** Remove outdated or unnecessary records

✔ 5. URL Routing System

Django URL routing maps web requests to appropriate views.
Functions:

- Defines application routes
- Connects frontend pages with backend logic
- Ensures smooth navigation between pages

Example:

- `/login/` → Login page
- `/home/` → Dashboard page

✔ 6. Session Management

Django handles session management to maintain user login state.
Features:

- Keeps users logged in during active session
- Protects restricted pages
- Logs out users securely

✔ 7. Data Validation and Security

The backend ensures secure and valid data processing.

Security Features:

- CSRF protection
- Password hashing
- Input validation
- Secure database queries

✔ 8. Admin Panel

Django provides a built-in admin panel for managing database records.
Features:

- Manage users and data easily
- Add or delete records
- Monitor system data in real-time

CHAPTER 11 – CODING & IMPLEMENTATION

The implementation of the Brainy Campus – Smart Academic Management System was carried out using the **Django framework with Python**, along with HTML, CSS, and SQLite database. The development process focused on building a modular, scalable, and user-friendly academic management system.

✔ 1. Project Setup

The first step was setting up the development environment.
Steps followed:

- Installed Python and Django framework
- Created a virtual environment
- Installed required dependencies

Started Django project using:
```
django-admin startproject brainycampus
```

Created application using:
```
python manage.py startapp myapp
```

✔ 2. Project Structure

The project was organized into two main parts:

- **Frontend:** HTML, CSS templates
- **Backend:** Django (Python logic + database handling)

Main folders include:

- models.py → Database structure
- views.py → Logic handling
- urls.py → Routing system
- templates/ → UI pages
- settings.py → Project configuration

✔ 3. Database Implementation

SQLite database was used for storing academic data.

Steps:

- Defined models in models.py
  Created migrations using:

  ```
  python manage.py makemigrations
  python manage.py migrate
  ```

- Connected models with Django admin panel

✔ 4. Backend Development

Backend logic was implemented using Django views.
Key functionalities:

- User authentication (login/logout)
- Dashboard rendering
- Data handling for modules
- CRUD operations for all academic records

Example:

- Login validation handled in `views.py`
- Home page displayed after successful login

✔ 5. Frontend Development

Frontend was developed using HTML and CSS.
Features:

- Login page
- Dashboard interface
- Module pages (Attendance, Assignment, Notes, etc.)
- Responsive and clean UI

✔ 6. Authentication System

Django's built-in authentication system was used.
Features:

- Secure login system
- Session management
- User role handling (Student/Admin)

✔ 7. Running the Project

The project is executed using the Django development server:
python manage.py runserver
After running, the system can be accessed via:
http://127.0.0.1:8000/

1. views.py

```python
# ================= LOGIN =================
def login_view(request):
    if request.user.is_authenticated:
        return redirect('home')

    if request.method == "POST":
        user = authenticate(
            request,
            username=request.POST.get('username'),
            password=request.POST.get('password')
        )

        if user:
            login(request, user)
            return redirect('home')

        return render(request, 'login.html', {'error': 'Invalid credentials'})

    return render(request, 'login.html')


# ================= HOME =================
@login_required(login_url='/login/')
def home_view(request):
    return render(request, 'home.html')


# ================= LOGOUT =================
def logout_view(request):
    logout(request)
    return redirect('/login/')
```

- login_view function
- home_view function

✔ 2. models.py

Shows database structure.

- Attendance model

```python
# ================= ATTENDANCE =================

class Attendance(models.Model):

    STATUS_CHOICES = [
        ('P', 'Present'),
        ('A', 'Absent'),
    ]

    student = models.ForeignKey(
        User,
        on_delete=models.CASCADE
    )

    class_name = models.CharField(max_length=50)

    subject = models.CharField(max_length=100)

    date = models.DateField()

    status = models.CharField(
        max_length=1,
        choices=STATUS_CHOICES
    )

    marked_by = models.ForeignKey(
        User,
        on_delete=models.SET_NULL,
        null=True,
        related_name='marked_attendance'
    )

    def __str__(self):
        return f"{self.student.username} - {self.subject} - {self.status}"
```

-

- Assignment model

```python
# ================= ASSIGNMENT =================
class Assignment(models.Model):

    teacher = models.ForeignKey(
        User,
        on_delete=models.CASCADE,
        null=True,
        blank=True
    )

    title = models.CharField(max_length=200)

    class_name = models.CharField(max_length=100)

    subject = models.CharField(max_length=100)

    description = models.TextField(blank=True)

    due_date = models.DateField()

    question_pdf = models.FileField(
        upload_to='assignments/',
        null=True,
        blank=True
    )

    status = models.CharField(
        max_length=20,
        default='Pending'
    )

    created_at = models.DateTimeField(auto_now_add=True)

    def is_expired(self):
        return date.today() > self.due_date

    def __str__(self):
        return self.title
```

-

- Notes model

```
# ================= NOTES =================
class Note(models.Model):

    user = models.ForeignKey(
        User,
        on_delete=models.CASCADE,
        related_name="notes"
    )

    title = models.CharField(max_length=200)

    content = models.TextField(blank=True, null=True)

    # NEW: PDF SUPPORT
    pdf_file = models.FileField(
        upload_to='notes_pdfs/',
        blank=True,
        null=True
    )

    created_at = models.DateTimeField(default=timezone.now)

    class Meta:
        ordering = ['-created_at']

    def __str__(self):
        return self.title
```

-

✔ 3. urls.py

Routing system.

```
brainycampus > myapp > urls.py
from django.urls import path, include
from . import views

urlpatterns = [

    # ================= AUTH =================
    path('', views.login_view, name='login'),
    path('login/', views.login_view, name='login'),
    path('home/', views.home_view, name='home'),
    path('logout/', views.logout_view, name='logout'),

    # ================= ATTENDANCE =================
    path('attendance/', views.attendance_view, name='attendance'),

    # ================= NOTES =================
    path('notes/', views.notes_view, name='notes'),
    path('add-note/', views.add_note, name='add_note'),
    path('delete-note/<int:id>/', views.delete_note, name='delete_note'),

    # ================= ASSIGNMENT =================
    path('assignments/', views.assignment_view, name='assignment'),
    path('submit-assignment/<int:assignment_id>/', views.submit_assignment, name='submit_assignm
    path('view-submissions/<int:assignment_id>/', views.view_submissions, name='view_submissions
    path('delete-assignment/<int:assignment_id>/', views.delete_assignment, name='delete_assignm

    # ================= TIMETABLE =================
    path('timetable/', views.timetable_view, name='timetable'),
    path('timetable/add/', views.add_timetable, name='add_timetable'),
    path('timetable/delete/<int:id>/', views.delete_timetable, name='delete_timetable'),
    path('timetable/edit/<int:id>/', views.edit_timetable, name='edit_timetable'),
    path('add-student/', views.add_student, name='add_student'),

    # ================= EXAM PLANNER =================
    path('exams/', include('myapp.exam_urls')),

    #======================Notebook===================#
    path('notebook/', views.notebook_view, name='notebook'),
    path('notebook/upload/', views.upload_material, name='upload_material'),
    path('delete-material/<int:pk>/', views.delete_material, name='delete_material'),

    #============ ADD STUDENT=============#
    path('students/add/', views.add_student, name='add_student'),
    path('students/', views.view_students, name='view_students'),
    path('students/edit/<int:pk>/', views.edit_student, name='edit_student'),
    path('students/delete/<int:pk>/', views.delete_student, name='delete_student'),

    #=================EVENTS================#
    path('events/', views.event_list, name='event_list'),
```

CHAPTER 12 – WORKING PROCESS

The Brainy Campus – Smart Academic Management System follows a structured workflow that allows users to access and manage academic activities in an efficient and systematic manner. The working process ensures smooth interaction between the user interface, backend system, and database.

✔ Step-by-Step Working Process

🔹 1. User Login

The process starts when the user opens the application and enters login credentials such as username and password.

🔹 2. Authentication

Django verifies the entered credentials using the authentication system.

- If credentials are correct → user is allowed to proceed
- If credentials are incorrect → access is denied

🔹 3. Dashboard Access

After successful login, the user is redirected to the dashboard. The dashboard acts as the main control center of the system where all modules are accessible.

🔹 4. Module Selection

The user selects any module such as:

- Attendance
- Assignment
- Notes
- Timetable
- Exam Planner
- Event Calendar

Each module performs a specific academic function.

🔹 5. Database Interaction

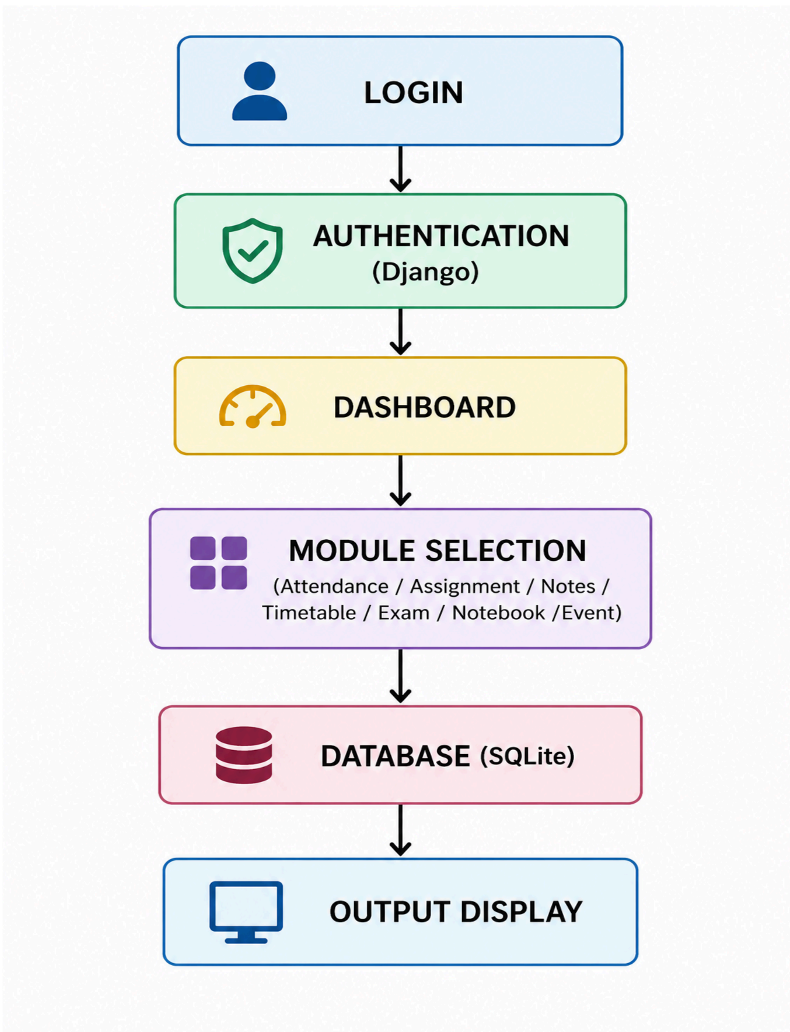
When a module is used:

- Data is sent to the backend (Django)
- Django processes the request
- Data is stored or retrieved from the SQLite database

🔹 6. Output Display

Finally, the processed data is displayed back to the user on the interface in a structured format.

FLOWCHART :

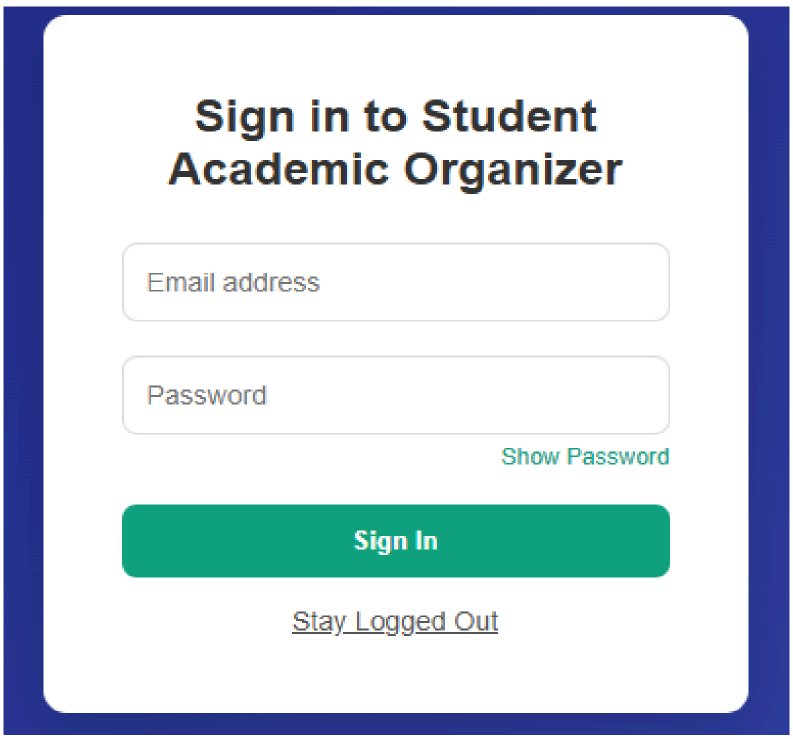


CHAPTER 13 – SCREENSHOTS & RESULTS

The Brainy Campus – Smart Academic Management System provides a clean and user-friendly interface for managing academic activities. This chapter presents the screenshots of the developed system along with their brief descriptions.

✔ 1. Login Page

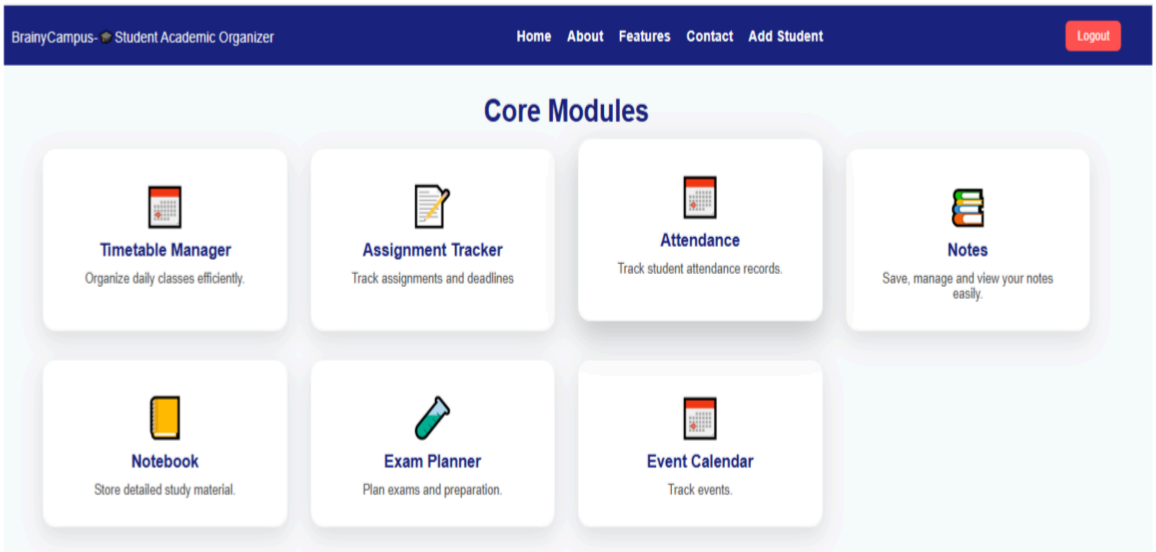
Fig – Login Page



The login page is the entry point of the system. Users enter their username and password to access the dashboard. It ensures secure authentication before granting access to the system.

✔ 2. Dashboard

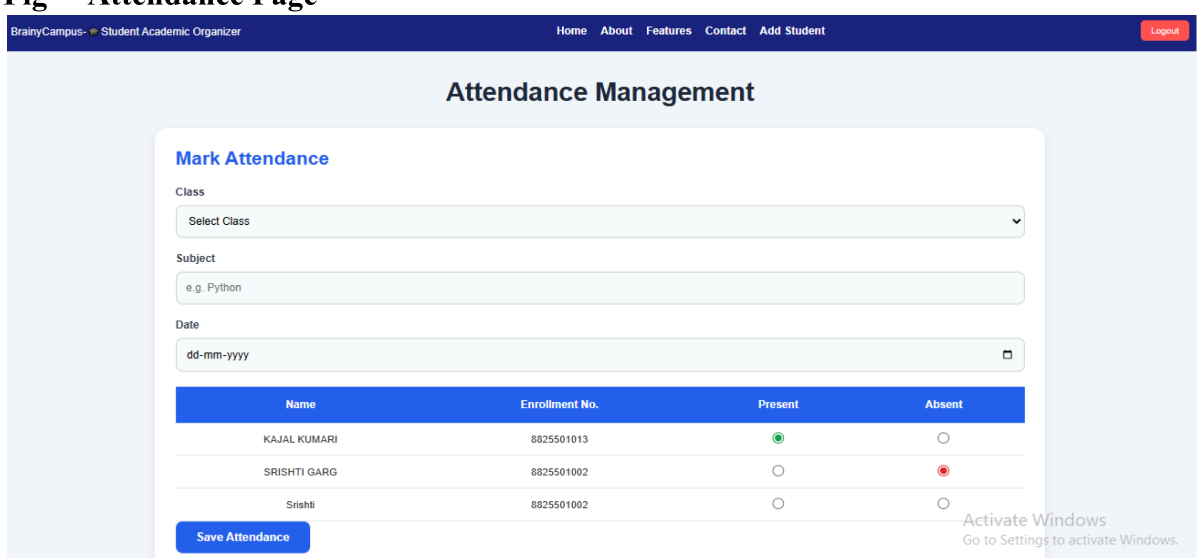
Fig – Dashboard Page



The dashboard is the main control panel of the Brainy Campus system. It provides quick access to all modules such as attendance, assignments, notes, timetable, and exam planner.

✔ 3. Attendance Module

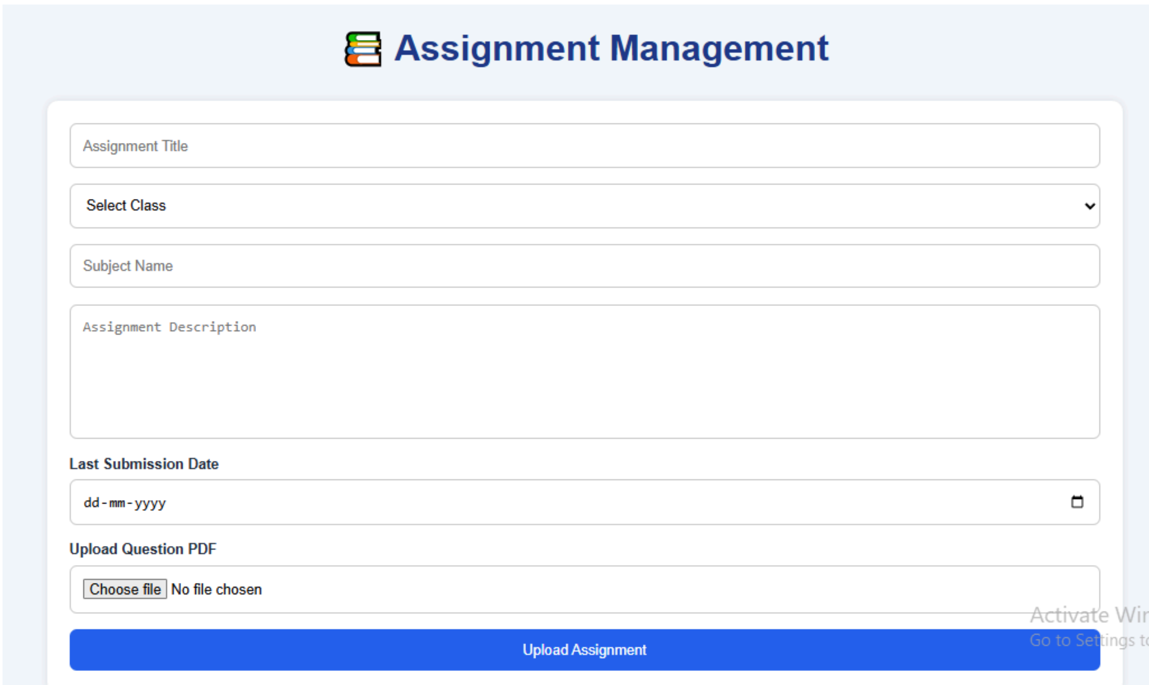
Fig – Attendance Page



This module allows students and teachers to manage and track attendance records. It also calculates attendance percentage automatically.

✔ 4. Assignment Module

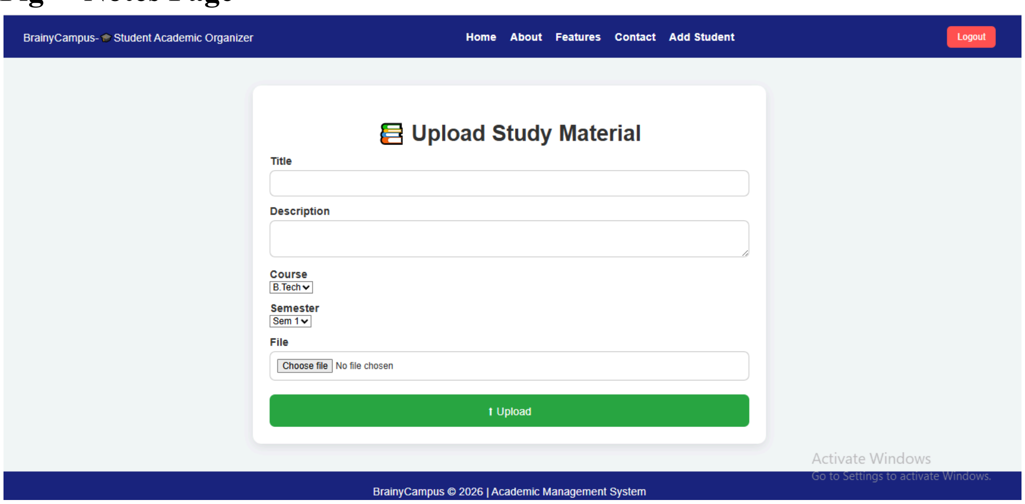
Fig – Assignment Page



The assignment module helps users track assignments, deadlines, and submission status. It improves academic organization and time management.

✔ 5. Notes Module

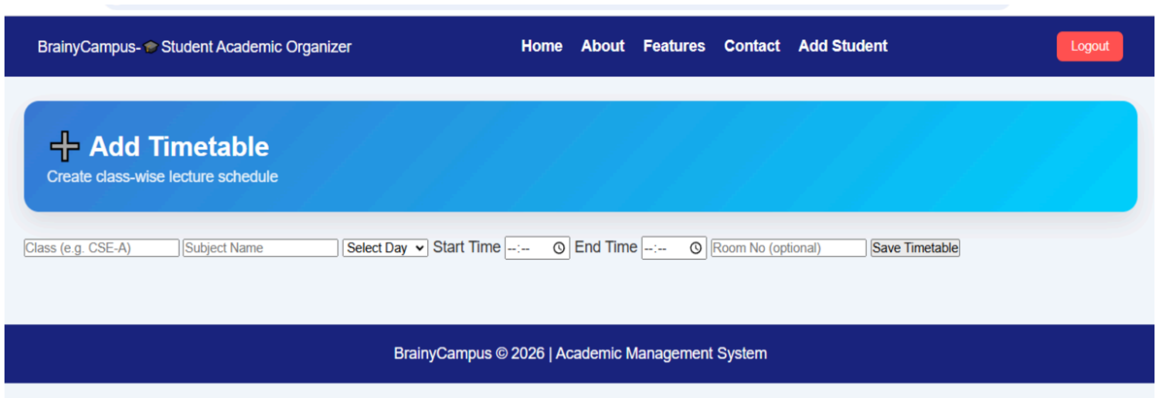
Fig – Notes Page



The notes module allows students to create, store, and manage study materials digitally for easy access and revision.

✔ 6. Timetable Module

Fig – Timetable Page



The timetable module displays daily class schedules in an organized manner, helping students manage their academic routine efficiently.

RESULTS

The Brainy Campus system was successfully developed and tested. The system provides:

- Secure login and authentication
- Centralized academic management
- Easy access to attendance, assignments, and notes
- Efficient timetable and exam planning
- User-friendly and responsive interface

The system improves productivity and reduces manual academic workload significantly.

CHAPTER 14 – ADVANTAGES

The Brainy Campus – Smart Academic Management System provides several advantages that improve the overall academic experience for students and educational institutions. It replaces traditional manual methods with a digital and automated platform, making academic management more efficient and reliable.

✔ 1. Centralized Academic Management

All academic activities such as attendance, assignments, notes, timetable, and exams are managed in a single platform, reducing confusion and improving organization.

✔ 2. Time Saving System

The system automates many manual tasks like attendance tracking and assignment management, saving a significant amount of time for both students and teachers.

✔ 3. Easy Accessibility

Users can access academic data anytime and from anywhere, making the system flexible and convenient to use.

✔ 4. Improved Productivity

By organizing academic tasks efficiently, students can focus more on studies rather than managing records manually.

✔ 5. Secure Data Management

The system uses Django authentication and database security to ensure that all academic data is protected from unauthorized access.

✔ 6. User-Friendly Interface

The system provides a simple, clean, and easy-to-use interface, making it suitable for all types of users.

✔ 7. Reduction in Paperwork

The digital system eliminates the need for physical records, reducing paperwork and making data management more eco-friendly.

✔ 8. Better Academic Tracking

Students can easily track attendance percentage, assignment deadlines, and exam schedules in one place.

CHAPTER 15 – LIMITATIONS

Although the Brainy Campus – Smart Academic Management System provides many useful features for academic management, there are still some limitations in the current version of the system. These limitations can be improved in future enhancements.

✔ 1. Internet Dependency

The system requires an active internet connection to access all features. Without internet connectivity, users cannot use the platform.

✔ 2. Basic User Interface

The current interface is simple and functional, but it can be further improved with more modern UI designs and advanced styling for better user experience.

✔ 3. Limited Scalability

The system is designed for small to medium-scale use. Handling a very large number of users may require additional optimization and stronger server support.

✔ 4. No Mobile Application

Currently, the system is only web-based. A mobile application version is not available, which limits accessibility for smartphone users.

✔ 5. Lack of Advanced Analytics

The system does not include advanced features like AI-based performance prediction or detailed academic analytics.

✔ 6. Basic Database System

SQLite is used as the database, which is suitable for small projects but not ideal for large institutional-level deployment.

✔ 7. No Real-Time Notification System

The system does not support advanced real-time notifications such as SMS or email alerts for deadlines and updates.

CHAPTER 16 – FUTURE SCOPE

The Brainy Campus – Smart Academic Management System has strong potential for future enhancement. With further development, the system can be upgraded into a more advanced and intelligent academic platform that supports modern educational requirements.

✔ 1. Mobile Application Development

In the future, a mobile application version of Brainy Campus can be developed for Android and iOS platforms. This will improve accessibility and allow students to manage academic activities anytime and anywhere.

✔ 2. AI-Based Features

Artificial Intelligence can be integrated to provide smart features such as:

- Personalized study recommendations
- Performance prediction
- Automated learning suggestions

✔ 3. Cloud Integration

The system can be deployed on cloud platforms to improve:

- Data storage capacity
- System scalability
- Remote accessibility
- Backup and recovery options

✔ 4. Real-Time Notification System

Future versions can include email, SMS, and push notifications for:

- Assignment deadlines
- Exam reminders
- Attendance alerts
- Event updates

✔ 5. Advanced Analytics Dashboard

An intelligent dashboard can be added to display:

- Student performance charts
- Attendance analytics
- Academic progress reports

✔ 6. Collaboration Features

The system can be enhanced with collaboration tools such as:

- Group discussions
- File sharing
- Live chat between students and teachers

✔ 7. Role-Based Expansion

More roles can be added in future such as:

- Faculty dashboard
- Admin control panel
- Department-wise management system

CHAPTER 17 – CONCLUSION

The Brainy Campus – Smart Academic Management System was successfully designed and developed to simplify and digitize academic activities for students and educational institutions. The system integrates multiple academic functions such as attendance management, assignment tracking, notes management, timetable scheduling, exam planning, and event management into a single centralized platform.

The project demonstrates the effective use of the Django framework for backend development along with Python, HTML, CSS, and SQLite for building a complete full-stack web application. The system follows a modular structure, which ensures better organization, maintainability, and scalability.

One of the major achievements of this project is the reduction of manual academic workload. Students can easily track their attendance, manage assignments, access notes, and plan their study schedule efficiently. Teachers can also manage academic records in a more structured and automated manner.

The implementation of authentication and role-based access ensures secure handling of user data. The system also improves productivity by providing quick access to all academic resources through a user-friendly dashboard.

Although the current system fulfills its objectives effectively, there is always scope for further enhancement by adding advanced features such as mobile application support, AI-based analytics, cloud integration, and real-time notifications.

In conclusion, Brainy Campus serves as an efficient and practical academic management solution that enhances the learning experience and supports the digital transformation of education systems.

CHAPTER 18 – REFERENCES

The following references and tools were used in the development of the **Brainy Campus – Smart Academic Management System**. The project has been successfully deployed and is accessible online.

✔ Documentation & Learning Resources

- Django Official Documentation
  https://docs.djangoproject.com/
- Python Official Documentation
  https://www.python.org/doc/
- MDN Web Docs (HTML, CSS, JavaScript)
  https://developer.mozilla.org/
- W3Schools Web Tutorials
  https://www.w3schools.com/

✔ Database Reference

- SQLite Official Documentation
  https://www.sqlite.org/docs.html

✔ Development Tools

- Visual Studio Code
  https://code.visualstudio.com/
- GitHub
  https://github.com/

✔ Deployment Platform

- Render (Cloud Hosting Platform)
  https://render.com/
- **Live Project URL:**
  https://brainycampus-5.onrender.com/home/

✔ Project Repository

GitHub link:
https://github.com/kajal345rajput/brainycampus

CHAPTER 19 – APPENDIX (SOURCE CODE)

The appendix section contains the important source code snippets of the **Brainy Campus – Smart Academic Management System** developed using Django framework. The system follows MVC architecture and uses Python for backend, HTML/CSS for frontend, and SQLite as the database.

✔ 1. Project Structure

```
brainycampus/
│
├── brainycampus/      (Project settings)
├── myapp/             (Main application)
├── templates/         (HTML files)
├── static/            (CSS & JS files)
├── db.sqlite3         (Database)
└── manage.py
```

✔ 2. settings.py (Important Configuration)

```
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'myapp',
]

DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.sqlite3',
        'NAME': BASE_DIR / 'db.sqlite3',
    }
}

STATIC_URL = '/static/'
```

✔ 3. models.py (Database Design)

```
from django.db import models
from django.contrib.auth.models import User

class Attendance(models.Model):
```

```
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    subject = models.CharField(max_length=100)
    total_classes = models.IntegerField(default=0)
    attended_classes = models.IntegerField(default=0)

class Assignment(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    title = models.CharField(max_length=200)
    due_date = models.DateField()
    status = models.BooleanField(default=False)

class Note(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    title = models.CharField(max_length=200)
    content = models.TextField()
```

✔ 4. views.py (Backend Logic)

```
from django.shortcuts import render, redirect
from django.contrib.auth import authenticate, login, logout

def login_view(request):
    if request.method == "POST":
        username = request.POST['username']
        password = request.POST['password']
        user = authenticate(request, username=username, password=password)
        if user:
            login(request, user)
            return redirect('home')
    return render(request, 'login.html')

def home_view(request):
    return render(request, 'home.html')

def logout_view(request):
    logout(request)
    return redirect('login')
```

✔ 5. urls.py (Routing)

Project urls.py

```
from django.contrib import admin
from django.urls import path, include
```

```
urlpatterns = [
    path('admin/', admin.site.urls),
    path('', include('myapp.urls')),
]
```

App urls.py

```
from django.urls import path
from .views import login_view, home_view, logout_view

urlpatterns = [
    path('', login_view, name='login'),
    path('home/', home_view, name='home'),
    path('logout/', logout_view, name='logout'),
]
```

✔ 6. HTML Sample (Frontend)

```
<h2>Login Page</h2>

<form method="POST">
    {% csrf_token %}
    <input type="text" name="username" placeholder="Username">
    <input type="password" name="password" placeholder="Password">
    <button type="submit">Login</button>
</form>
```

✔ 7. Database Used

- SQLite (`db.sqlite3`)
- Tables:
  - User
  - Attendance
  - Assignment
  - Notes
  - Timetable
  - Events

✔ 8. Deployment

The project is deployed on a cloud platform and is accessible online.
Live Project URL:
https://brainycampus-5.onrender.com/home/